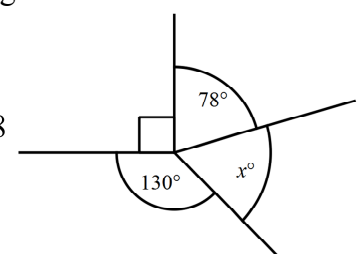
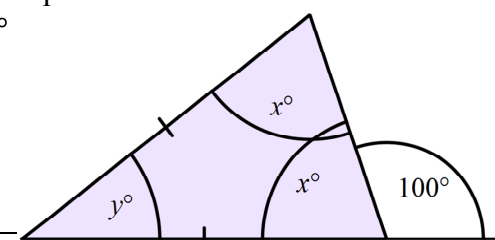


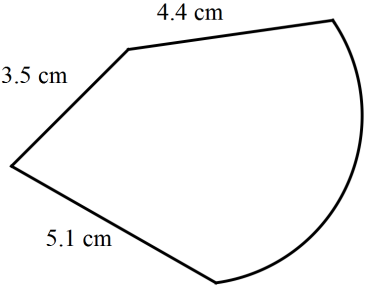
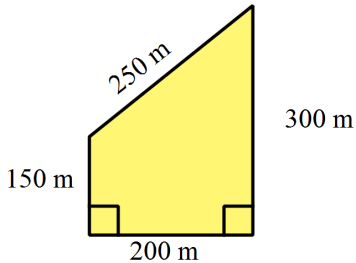


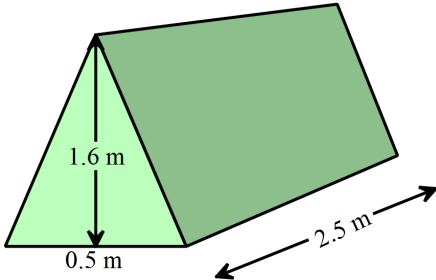
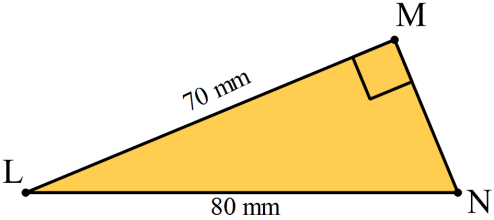
Western Mathematics
2013 Year 10
Half Yearly Examination
Mathematics Course
Solutions

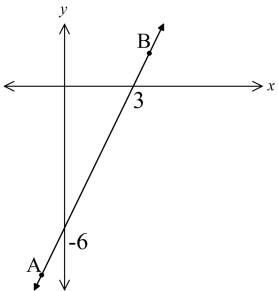
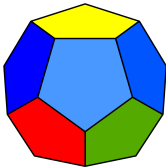
Short Answer Questions

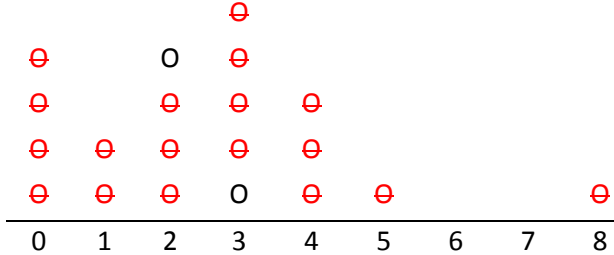
1.	$8 \times (15 - 3) = 8 \times 12 = 96$
2.	What is $\frac{2}{5}$ of 45kg = $\frac{2}{5} \times \frac{45}{1} = 18$ kg
3.	$1.2 - 0.35 =$ $\begin{array}{r} 1.20 \\ - 0.35 \\ \hline 0.85 \end{array}$
4.	Commission = 6% of \$2 400 $= \frac{6}{100} \times \frac{2400}{1}$ $= 24 \times 6$ $= \$144$
5.	$I = PRN$ $= 600 \times 0.07 \times 2$ $= 42.00 \times 2$ $= \$84.00$

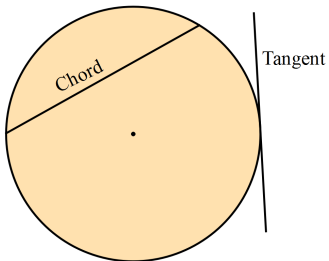
6.	Loss = $60000 - 45000 = 15\,000$ Percentage loss = $\frac{15000}{60000} = \frac{15}{60} \times 100$ $= \frac{1}{4} \times 100$ $= 25\%$
7.	$4.5 \text{ kilograms} : 750 \text{ grams} = 4500 \text{ g} : 750 \text{ g}$ $= 450 : 75$ $= 18 : 3$ $= 6 : 1$
8.	What is the value of x in the diagram? $\begin{aligned} x + 130 + 78 + 90 &= 360 \\ x + 298 &= 360 \\ x &= 360 - 298 \\ &= 62 \end{aligned}$ 
9.	The base angles x° are equal. $\begin{aligned} x^\circ &= 180 - 100 = 80^\circ \\ y^\circ &= 180 - 2 \times 80 \\ &= 180 - 160 \\ &= 20^\circ \end{aligned}$ 

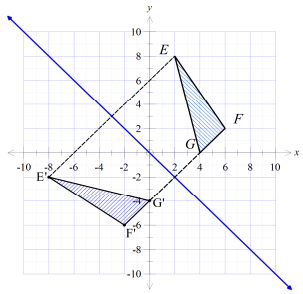
10.	Property	Yes /No
	All sides equal	Y
	All angles equal	N
	Diagonals are equal	N
	Diagonals intersect at right angles	Y
	Axes of symmetry	Y (2)
The name is a RHOMBUS		
11.	Volume = $10 \times 375 \text{ ml}$ $= 3750 \text{ ml}$ $= 3.75 \text{ litres}$	
12.	Perimeter = $4.4 + 3.5 + 5.1 + 13.0 + C = 20.4$ $C = 20.4 - 13.0$ $= 7.4 \text{ cm}$ 	
13.	$A = \frac{1}{2} h(a + b)$ $= \frac{1}{2} \times 200 (150 + 300)$ $= 100 \times 450$ $= 45\,000 \text{ m}^2$ 	

14.	$V = Ah$ $A = \frac{1}{2} \times 0.5 \times 1.6$ $= 0.25 \times 1.6$ $= 0.4$ $V = 0.4 \times 2.5$ $= 1.00 \text{ m}^3$ 
15.	$MN^2 = 80^2 - 70^2$ $= 6400 - 4900$ $= 1500$ $MN = \sqrt{1500} \text{ mm}$ 
16.	$13m^2 - 5m - 7m^2 - 8m = 6m^2 - 13m$
17.	When $x = 2, y = 5$ and $z = -4$ $\frac{2z^2 - x}{3y} = \frac{2 \times (-4)^2 - 2}{3 \times 5}$ $= \frac{2 \times 16 - 2}{15}$ $= \frac{30}{15}$ $= 2$
18.	$2d(2d + c) - 3c(2d - c) = 4d^2 + 2dc - 6dc + 3c^2$ $= 4d^2 - 4dc + 3c^2$

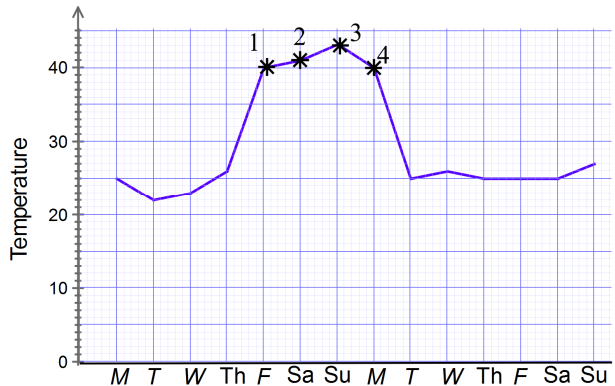
19.	$m = \text{rise/run}$ $= \frac{6}{3}$ $= 2$ OR $m = \frac{-6-0}{0-3}$ $= \frac{-6}{-3}$ $= 2$ 
20.	$4^{-3} = \frac{1}{4^3} = \frac{1}{64}$
21.	Solve $8x + 7 = 22 - 2x$ $10x + 7 = 22$ $10x = 15$ $x = \frac{15}{10}$ $x = 1.5$
22.	Number red = $12 - 2 - 2 - 5 = 3$ Want probability that a red side finishes facing up. $P(\text{red}) = \frac{3}{12}$ $= \frac{1}{4}$ 
23.	Dance and Games are both 60° which is $\frac{1}{6}$ of his time.

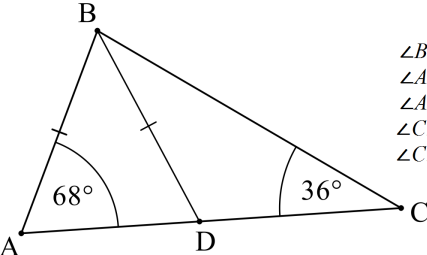
24.	James spent 6 hours a week playing sport = $\frac{1}{4}$ of total. Total = $6 \times 4 = 24$ hours Reading is 150° Reading is $\frac{150}{360}$ of total <i>hours</i> = $\frac{15}{36} = \frac{5}{12}$ Time reading = $\frac{5}{12} \times 24$ = 10 hours
25.	From 20 scores median between 10 th and 11 th so between 2 and 3. Median = 2.5 

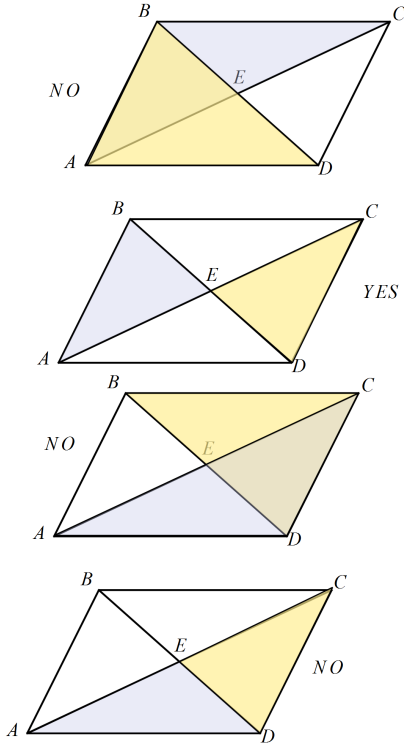
Multiple Choice Questions		
26.	$35\% = \frac{35}{100} = \frac{7}{20}$	D
27.	$\frac{2}{3} \div \frac{3}{4} = \frac{2}{3} \times \frac{4}{3}$	B
28.	Distance travelled in race = $6 \times 75 = 450$ km. Speed = Dist/Time Time = Dist/Speed $= \frac{450}{150}$ hours $= 3$ hours	D
29.	Interest = $900 \times 0.06 \times 3 = \162 Amount repaid = $900 + 162 = \$1062$	C
30.	40 hours normal plus 8 hours at time $\frac{1}{2}$ $= 40 + 12 = 52$ hours at normal rates Pay = $\$28 \times 52 = \1456	C
31.	$2g + 110 = 180$ (cointerior angles) $2g = 70$ $g = \frac{70}{2} = 35$	B
32.		D

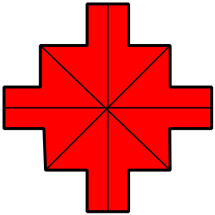
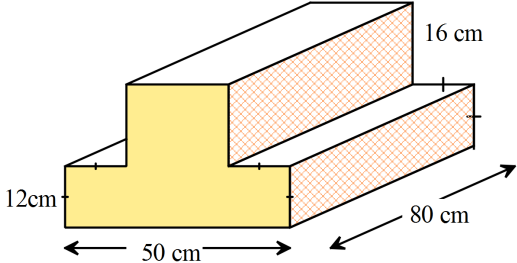
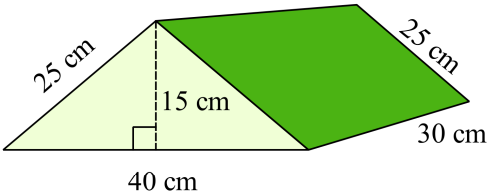
33.		A
34.	$2x + 3x + 90 = 180$ $5x = 90$ $x = \frac{90}{5}$ $= 18$	B
35.	$y = 360 - (72 + 80 + 85)$ $= 360 - 237$ $= 123^\circ$	C
36.	$A = \pi r^2$ $7200 = \pi \times r^2$ $r^2 = \frac{7200}{\pi}$ $r^2 \approx \frac{2292}{\pi}$ $r \approx \sqrt{2292}$ $\approx 47.87..$ $d \approx 95.75$ $d = 96 \text{ m (nearest metre)}$ Or by testing the four alternative answers.	D
37.	Diagonals of rhombus $x = 35 - 2 \times 7 = 21$ $y = 30 - 2 \times 6 = 18$ Area Rhombus = $\frac{1}{2} \times 21 \times 18 = 189 \text{ m}^2$ Area courtyard = $35 \times 30 - 189$ $= 1050 - 189$ $= 861 \text{ m}^2$	D

38.	$V = \pi r^2 h$ $= \pi \times 8^2 \times 3.2$ $= 643.3981$ $= 643 \text{ m}^3 \text{ (nearest m}^3\text{)}$	A
39.	$B^2 = 1.8^2 + 1.3^2$ $= 4.93$ $B = \sqrt{4.93}$ $= 2.2 \text{ m}$	C
40.	$\sin \theta = \frac{O}{H} = \frac{24}{25}$	C
41.	$C = \frac{5}{9}(F - 32)$ $C = \frac{5}{9}(-22 - 32)$ $C = \frac{5}{9}(-54)$ $C = -30$	B
42.	$3s - 4st - 5s(3 - 2t) = 3s - 4st - 15s + 10st$ $= -12s + 6st$ $= 6st - 12s$	A
43.	$\text{Midpoint} = \left(\frac{-5 + 5}{2}, \frac{-5 + 4}{2} \right)$ $= \left(\frac{0}{2}, \frac{-1}{2} \right)$ $= \left(0, -\frac{1}{2} \right)$	B
44.	$\frac{u^4 \times u^8}{u^2 \times u^4} = \frac{u^{4+8}}{u^{2+4}}$ $= \frac{u^{12}}{u^6}$ $= u^{12-6}$ $= u^6$	D

45.	$\frac{6 - 2m}{5} = 4m + 3$ $\cancel{5} \times (6 - 2m) = 5 \times (4m + 3)$ $6 - 2m = 5(4m + 3)$	A																		
46.	Choosing 3 males and 4 females from her class of 28 students is a sample of the class.	B																		
47.	 4 days at \$10 per day = \$40	D																		
48.	<table border="1" data-bbox="1339 906 1771 1171"> <thead> <tr> <th>Number of Calls (x)</th> <th>Frequency (f)</th> <th>fx</th> </tr> </thead> <tbody> <tr> <td>6</td> <td>3</td> <td>18</td> </tr> <tr> <td>7</td> <td>10</td> <td>70</td> </tr> <tr> <td>8</td> <td>11</td> <td>88</td> </tr> <tr> <td>9</td> <td>4</td> <td>36</td> </tr> <tr> <td>10</td> <td>2</td> <td>20</td> </tr> </tbody> </table> $\Sigma f = 30 \quad \Sigma fx = 232$ $\text{Mean} = \frac{232}{30} = 7.733 = 7.7 \text{ (1 dec place)}$	Number of Calls (x)	Frequency (f)	fx	6	3	18	7	10	70	8	11	88	9	4	36	10	2	20	A
Number of Calls (x)	Frequency (f)	fx																		
6	3	18																		
7	10	70																		
8	11	88																		
9	4	36																		
10	2	20																		
49.	76 and 98 both occur three times so are both modes.	A																		
50.	Range = $100 - 67 = 33$	C																		

51.	$\begin{array}{r} 7\ 4\ 5\ - \\ 2\ 6\ 8 \\ \hline 4\ 7\ 7 \end{array}$ $\triangle = 7$ and $\square = 6$	A
52.	$\begin{aligned} I &= PRN \\ &= 6500 \times 0.0425 \times 1.5 \\ &= 414.375 \\ &= \$414.38 \end{aligned}$	A
53.	$\begin{aligned} \text{Price by paying off} &= 342 + 70 \times 24 \\ &= 342 + 1680 \\ &= 2022 \\ \text{Extra paid} &= 2022 - 1600 \\ &= \$422 \end{aligned}$	B
54.	$\begin{aligned} A &= P(1 + R)^N \\ &= 12000(1.07)^4 \\ &= 15729.55212 \\ &= \$15\ 729.55 \end{aligned}$	D
55.	Half is \$1250. Trent receives $\frac{2}{5} \times 1250 = \\500	A
56.	 $\angle BDA = 68$ $\angle ABD = 180 - 2 \times 68 = 44^\circ$ $\angle ABC = 180 - 68 - 36 = 76^\circ$ $\angle CBD = \angle ABC - \angle ABD$ $\angle CBD = 76 - 44 = 32^\circ$	A

57.	 B. $\triangle BAE$ and $\triangle DCE$.	B
58.	$\begin{aligned} \text{Angle sum} &= (n - 2) \times 180 \\ &= (6 - 2) \times 180 \\ &= 4 \times 180 \\ &= 720^\circ \\ 4x + 180 &= 720 \\ 4x &= 540 \\ x &= \frac{540}{4} = 135^\circ \end{aligned}$	C

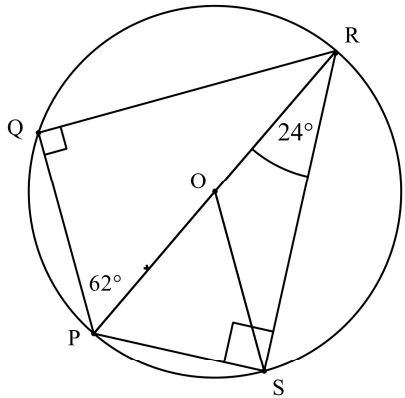
59.	 4 axes of line symmetry Rotational Symmetry order 4 I is false II is true	D
60.	 Area = $50 \times 12 + 26 \times 16 = 1016 \text{ cm}^2$ Volume = $Ah = 1016 \times 80 = 81\,280 \text{ cm}^3$	D
61.	 $A = 25 \times 30 \times 2 + 40 \times 30 + 2 \times \frac{1}{2} \times 40 \times 15$ $= 1500 + 1200 + 600$ $= 3300 \text{ cm}^2$	C

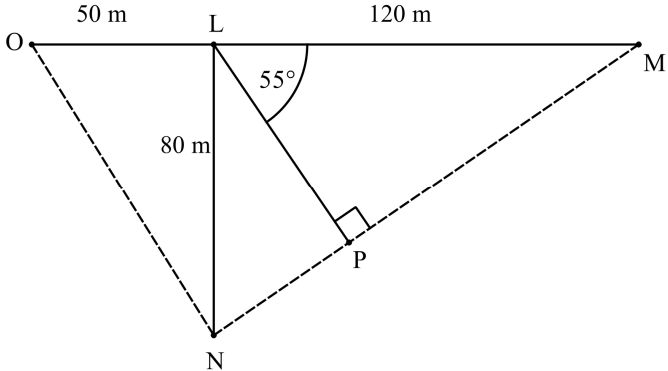
62.	$\begin{aligned} \text{Area End} &= \pi r^2 \\ &= \pi \times 10^2 \\ &= 314 \\ \text{Area Curve} &= 2\pi r h \\ &= 2 \times \pi \times 10 \times 40 \\ &= 2513 \\ \text{Surface Area} &= 314 + 2513 \\ &= 2827 \text{ cm}^2 \end{aligned}$	C
63.	$\begin{aligned} \cos \beta &= \frac{3}{5} \\ \beta &= \cos^{-1} \left(\frac{3}{5} \right) \\ \beta &= 53.130 = 53^\circ \text{ (nearest degree)} \end{aligned}$	C
64.	$\begin{aligned} \sin 47 &= \frac{k}{35.2} \\ k &= 35.2 \times \sin 47 \\ k &= 25.7 \text{ cm} \end{aligned}$	A
65.	$\begin{aligned} \tan A &= \frac{190}{110} \\ A &= \tan^{-1} \left(\frac{190}{110} \right) \\ A &= 60^\circ \end{aligned}$	B
66.	$12ax^2 - 16a^2x = 4ax(3x - 4a)$ So the first three are all factors but D is not.	D
67.	Gradient of -4 and passes through the point (1, -5) $\begin{aligned} y &= -4x + b \\ -5 &= -4 \times 1 + b \\ -5 &= -4 + b \\ b &= -1 \\ y &= -4x - 1 \end{aligned}$ OR $\begin{aligned} y - y_1 &= m(x - x_1) \\ y - (-5) &= -4(x - 1) \\ y + 5 &= -4x + 4 \\ y &= -4x - 1 \end{aligned}$	C

68.	$A(-2, 5)$ $B(4, -3)$ $AB^2 = 6^2 + 8^2$ $AB^2 = 100$ $AB = \sqrt{100} = 10$	D
69.	$\frac{6a^2z^3}{5bx^2} \times \frac{15b^3x}{12a^4z^3} = \frac{\cancel{6}a^2z^3}{\cancel{5}bx^2} \times \frac{\cancel{15}b^3x\cancel{3}}{\cancel{12}a^4z^3\cancel{2}}$ $= \frac{3a^2b^3xz^3}{2a^4bx^2z^3}$ $= \frac{3b^2}{2a^2x}$	C
70.	$\frac{4x}{3} - 8 = 2x - 7$ $\cancel{3} \times \frac{4x}{\cancel{3}} - 8 \times 3 = 3 \times 2x - 3 \times 7$ $4x - 24 = 6x - 21$ $-2x - 24 = -21$ $-2x = 3$ $x = \frac{3}{-2} = -1\frac{1}{2}$	A

Ages of Emattogah Cricket Players		
1	4	5
2	0	2
3	1	3
4	2	6
5	3	

71.	Counting the leaves, there are 23.	B
72.	From 23 the middle is the 12 th leaving 11 above and 11 below. The 12 th score is 29	C
73.	From the upper 11 scores the middle is the 6 th of these leaving 5 above and 5 below, 6 th from top is the 18 th score which is 35.	B
74.	It is both Bimodal (2 and 8) and symmetrical.	B
75.	From observation mean = 5, median = 5, modes = 2 and 8 and range = 9 - 1 = 8 Mean and median are equal.	C

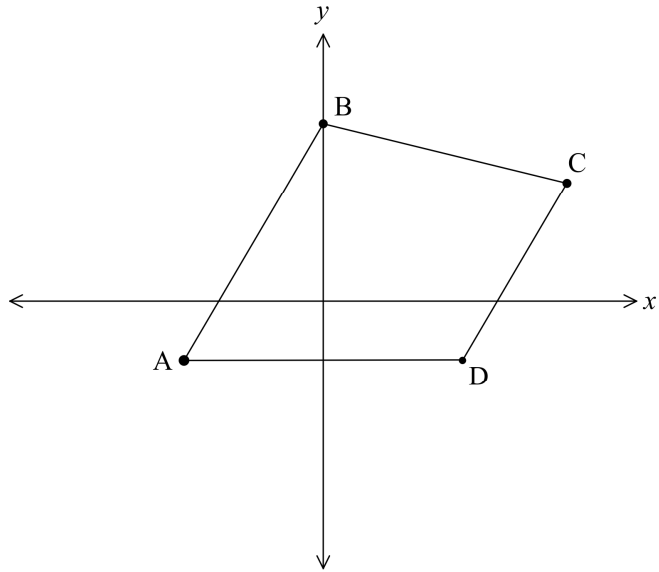
Longer Answer Questions		
76. a)	\$1 grows to \$1.5869 \$7500 grows to $1.5869 \times 7500 = \$11\,901.75$	1 mark
b)	If \$8000 grows to \$10100 Then \$1 grows to $10100 \div 8000 = 1.2625$ From table on the 4 year row 1.2625 is under 6% Required interest rate is 6% p.a.	1 mark
c)	$A = P(1 + R)^N$ $= 7500(1.08)^{10}$ $= \$16\,191.94$ Extra Earned = $\$16\,191.94 - \$11\,901.75$ $= \$4\,290.19$	1 mark
77.	 $\angle OSR = \angle ORS = 24^\circ$ ($\triangle ORS$ is isosceles – equal radii) $\angle OSP = 90 - 24 = 66^\circ$ (Complementary angles) $\angle SPO = \angle OSP = 66^\circ$ ($\triangle OSP$ is isosceles – equal radii) $\angle SPQ = 62^\circ + 66^\circ = 128^\circ$ (Adjacent angles)	3 marks total 3 marks for complete answer with reasons 2 or 1 mark for partial attempts

78. a)	Area of curved surface = $2 \pi r h$ $= 2 \times \pi \times 15 \times 10$ $= 942.4 \text{ m}^2$ Amount of paint = $942.4 \div 25$ $= 37.7 \text{ litres}$	2 marks total 1 for area 1 for amount of paint
b)	$V = \pi r^2 h$ $= \pi \times 15^2 \times 10$ $= 7068.6 \text{ m}^2$ $C = 7068.6 \times 1000$ $= 7\,069\,000 \text{ litres (nearest 1000 litres)}$	1 mark
79. a)	 $ON^2 = 50^2 + 80^2$ $= 8900$ $ON = \sqrt{8900} = 94 \text{ m (nearest metre)}$	1 mark

b)	$\tan \angle LON = \frac{80}{50}$ $\angle LON = \tan^{-1}\left(\frac{80}{50}\right)$ $= 57.9$ $= 58^\circ$ Bearing = $90 + 58 = 148^\circ$	2 marks total 1 for angle 1 for bearing
c)	$\sin 55^\circ = \frac{PM}{120}$ $PM = 120 \times \sin 55^\circ$ $= 98.29$ $= 98 \text{ m (nearest metre)}$	1 mark
80. a)	4, 6, 8 and 12 are on black segments. $P(\text{Black and even}) = \frac{4}{12} = \frac{1}{3}$	1 mark
b)	1, 2, 3, and 4 are less than 5. 4, 5, 6, 7, 8 and 12 are on black segments. 4 is on both lists so only count it once, so required numbers are 1, 2, 3, 4, 5, 6, 7, 8 and 12. $P(\text{less 5 or black or both}) = \frac{9}{12} = \frac{3}{4}$	1 mark
81 a)	Sum of Scores = $61 + 62 + 63 \times 2 + 64 \times 3 + 65 \times 5 + 66 \times 7 + 67 \times 11 + 68 \times 12 + 69 \times 7 + 70 \times 6 + 71 \times 3$ $= 3897$ Mean = $\frac{3879}{58}$ $= 67.2 \text{ (1 dec place)}$	1 mark

b)	Both Rounds have skewed distributions but in reverse to one another. Round 1 is skewed toward the higher scores and round 2 is skewed toward the lower scores. Round 1 would have a higher mean and median than Round 2.	2 marks total 2 marks for a description that correctly mentions different skew and relative value of the means/medians. 1 if one of these is mentioned correctly.
82. a)	$\frac{a}{6} + \frac{4}{3a} = \frac{a}{6} \times \frac{a}{a} + \frac{4}{3a} \times \frac{2}{2}$ $= \frac{a^2}{6a} + \frac{8}{6a}$ $= \frac{a^2 + 8}{6a}$	2 marks total 2 marks for correct answer 1 for some correct working with a minor error

b)	$2x - \frac{5-x}{3} = \frac{3x}{2}$ $6 \times 2x - \frac{6(5-x)}{3} = \frac{6 \times 3x}{2}$ $12x - 2(5-x) = 9x$ $12x - 10 + 2x = 9x$ $5x - 10 = 0$ $5x = 10$ $x = \frac{10}{5}$ $x = 2$	2 marks total 2 marks for correct solution 1 for some correct working with an error
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83.	 AB and CD appear parallel. $m_{AB} = \frac{6 - -2}{0 - -4}$ $= \frac{8}{4}$ $= 2$ $m_{CD} = \frac{-2 - 4}{4 - 7}$ $= -\frac{6}{-3}$ $= 2$ $\therefore AB \parallel CD$ $\therefore$ ABCD is a trapezium	3 marks total 1 gradient 1 gradient 1 conclusion
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High School

Half - Yearly Exam

Mathematics Course

Multiple Choice Section Answer Sheet

Name MARKING SHEET

Teacher _____

Completely fill the response oval representing the most correct answer.

26. A ☐ B ☐ C ☐ D ☒
27. A ☐ B ☒ C ☐ D ☐
28. A ☐ B ☐ C ☐ D ☒
29. A ☐ B ☐ C ☒ D ☐
30. A ☐ B ☐ C ☒ D ☐
31. A ☐ B ☒ C ☐ D ☐
32. A ☐ B ☐ C ☐ D ☒
33. A ☒ B ☐ C ☐ D ☐
34. A ☐ B ☒ C ☐ D ☐
35. A ☐ B ☐ C ☒ D ☐
36. A ☐ B ☐ C ☐ D ☒
37. A ☐ B ☐ C ☐ D ☒
38. A ☒ B ☐ C ☐ D ☐
39. A ☐ B ☐ C ☒ D ☐
40. A ☐ B ☐ C ☒ D ☐
41. A ☐ B ☒ C ☐ D ☐
42. A ☒ B ☐ C ☐ D ☐
43. A ☐ B ☒ C ☐ D ☐
44. A ☐ B ☐ C ☐ D ☒
45. A ☒ B ☐ C ☐ D ☐
46. A ☐ B ☒ C ☐ D ☐
47. A ☐ B ☐ C ☐ D ☒
48. A ☒ B ☐ C ☐ D ☐
49. A ☒ B ☐ C ☐ D ☐
50. A ☐ B ☐ C ☒ D ☐

51. A ☒ B ☐ C ☐ D ☐
52. A ☒ B ☐ C ☐ D ☐
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63. A ☐ B ☐ C ☒ D ☐
64. A ☒ B ☐ C ☐ D ☐
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66. A ☐ B ☐ C ☐ D ☒
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68. A ☐ B ☐ C ☐ D ☒
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70. A ☒ B ☐ C ☐ D ☐
71. A ☐ B ☒ C ☐ D ☐
72. A ☐ B ☐ C ☒ D ☐
73. A ☐ B ☒ C ☐ D ☐
74. A ☐ B ☒ C ☐ D ☐
75. A ☐ B ☐ C ☒ D ☐

